

Healthcare Risk Analytics Dashboard: Insights Report

Executive Summary

Healthcare organizations face increasing pressure to improve patient outcomes while managing operational efficiency. Leveraging data analytics to identify risk factors, monitor disease prevalence, and understand patient behaviors can support proactive healthcare interventions and evidence-based decision-making.

This project analyzed approximately 30,000 patient records using Power BI, Power Query, DAX, and Excel to evaluate patient demographics, clinical indicators, lifestyle behaviors, risk profiles, and hospital operational metrics. The analysis resulted in the development of an interactive Healthcare Risk Analytics Dashboard designed to support healthcare stakeholders in identifying high-risk populations and understanding key drivers of patient health outcomes.

The findings revealed significant relationships between clinical and lifestyle indicators, highlighting the role of diet, physical activity, stress management, and clinical monitoring in patient risk management.

Business Objective

The objective of this project was to transform raw healthcare data into actionable insights that support:

- Patient risk identification and stratification.
- Disease prevalence monitoring.
- Lifestyle risk assessment.
- Clinical outcome evaluation.
- Hospital operational performance analysis.
- Data-driven healthcare decision-making.

Dataset Overview

The dataset consisted of approximately 30,000 patient records containing demographic, clinical, lifestyle, and operational healthcare information.

Key Data Categories

Demographic Variables

- Age
- Gender

Clinical Indicators

- Blood Pressure

- Glucose
- BMI
- Cholesterol
- Triglycerides
- HbA1c
- Oxygen Saturation

Lifestyle Indicators

- Smoking Status
- Alcohol Consumption
- Physical Activity
- Diet Score
- Stress Level
- Sleep Hours

Operational Indicators

- Length of Stay (LOS)

Medical Information

- Medical Condition
- Family History

Methodology

The project followed a structured analytics workflow:

1. Data Cleaning and Preparation
2. Data Modeling
3. Feature Engineering
4. Risk Scoring Development
5. Correlation Analysis
6. Dashboard Development
7. Insight Generation

Data Preparation

Data preparation activities included:

- Handling missing values.
- Correcting data types.
- Removing noise variables.
- Creating age groups and BMI categories.
- Developing custom risk segmentation models.
- Creating analytical reference tables for advanced visualizations.

Risk Scoring Framework

To support risk stratification, two custom scoring models were developed.

Cardiac Risk Score

The cardiac risk model incorporated:

- Blood Pressure
- Cholesterol
- Triglycerides
- HbA1c
- Smoking Status
- Alcohol Consumption
- Family History

Patients were classified into:

- Low Risk
- Moderate Risk
- High Risk

Lifestyle Risk Score

Lifestyle risk assessment was based on:

- Physical Activity
- Diet Score

- Stress Level
- Sleep Hours

Patients were categorized into:

- Healthy Lifestyle
- Moderate Lifestyle Risk
- High Lifestyle Risk

Key Findings

1. Disease Burden Remains Significant

The dashboard revealed a substantial proportion of patients living with chronic health conditions, including diabetes, hypertension, and cardiovascular-related illnesses.

This highlights the continued need for preventive healthcare interventions and risk monitoring programs.

Business Implication

Healthcare providers should prioritize preventive care strategies and patient education initiatives aimed at reducing chronic disease burden.

2. Glucose and HbA1c Exhibit the Strongest Relationship

Correlation analysis identified the strongest positive relationship between glucose and HbA1c levels ($r = 0.62$).

This finding aligns with established clinical evidence, as HbA1c reflects long-term blood glucose control.

Business Implication

Routine monitoring of glucose levels can serve as an early indicator of poor glycemic control and support timely intervention.

3. Lifestyle Behaviors Influence Health Outcomes

Several lifestyle variables demonstrated meaningful relationships with health indicators.

Key Observations

- Higher physical activity levels were associated with lower BMI values.
- Better diet scores were associated with lower glucose levels.

- Better diet quality was linked to lower HbA1c values.
- Higher stress levels were associated with elevated blood pressure.

Business Implication

Healthcare organizations should invest in lifestyle intervention programs that promote healthy eating, physical activity, stress management, and adequate sleep.

4. Age Contributes to Elevated Clinical Risk

Analysis showed a positive relationship between age and blood pressure.

Older patients generally exhibited higher cardiovascular risk indicators than younger populations.

Business Implication

Age-specific risk screening programs may improve early detection and prevention efforts.

5. High-Risk Patients Experience Greater Operational Impact

Patients categorized as high-risk generally demonstrated longer hospital stays compared to lower-risk groups.

This indicates that elevated clinical risk is associated with increased healthcare resource utilization.

Business Implication

Risk-based patient management strategies may help optimize healthcare resource allocation and improve operational efficiency.

6. Length of Stay Varies Across Patient Groups

Length of Stay analysis revealed variations across:

- Medical conditions
- Age groups
- BMI categories
- Risk categories

These differences provide valuable insight into patient complexity and healthcare utilization patterns.

Business Implication

Healthcare administrators can leverage LOS insights to improve discharge planning and resource management.

Correlation Analysis Highlights

The strongest relationships identified in the dataset included:

Relationship	Correlation
Glucose ↔ HbA1c	0.62
Physical Activity ↔ Diet Score	0.47
HbA1c ↔ Diet Score	-0.35
Physical Activity ↔ BMI	-0.35
Diet Score ↔ Glucose	-0.31
Age ↔ Blood Pressure	0.30

Interpretation

The findings suggest that lifestyle behaviors play a measurable role in influencing patient health outcomes and risk profiles.

Healthcare organizations can leverage these insights to design targeted wellness and preventive care initiatives.

Recommendations

Based on the findings, the following recommendations are proposed:

Clinical Recommendations

- Strengthen diabetes monitoring programs.
- Enhance cardiovascular risk screening.
- Implement early intervention strategies for high-risk patients.

Lifestyle Recommendations

- Promote physical activity programs.
- Encourage healthy dietary habits.
- Introduce stress management initiatives.
- Support patient wellness education.

Operational Recommendations

- Utilize risk stratification to prioritize care delivery.
- Monitor high-risk patients more closely.
- Optimize resource planning using Length of Stay insights.

Conclusion

The Healthcare Risk Analytics Dashboard demonstrates how healthcare data can be transformed into actionable intelligence through business intelligence and analytics.

By combining clinical indicators, lifestyle behaviors, risk scoring methodologies, and operational performance metrics, the project provides a comprehensive view of patient health and healthcare utilization.

The analysis highlights the importance of proactive risk management, preventive healthcare strategies, and data-driven decision-making in improving patient outcomes and operational efficiency.

Ultimately, the project illustrates the value of analytics as a tool for identifying risk, uncovering patterns, and supporting better healthcare decisions.